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# STM32Cube Ecosystem

# STM32Cube Framework

Helping developers releasing their creativity



## Develop faster

With STM32 code generation tools, IDE and production-ready examples

## Differentiate more

With optimized drivers leveraging all STM32 smart features

## Save costs

With Free and permissive production licenses

**1.5M+**

Active developer

**76% CSAT**

at professional engineers

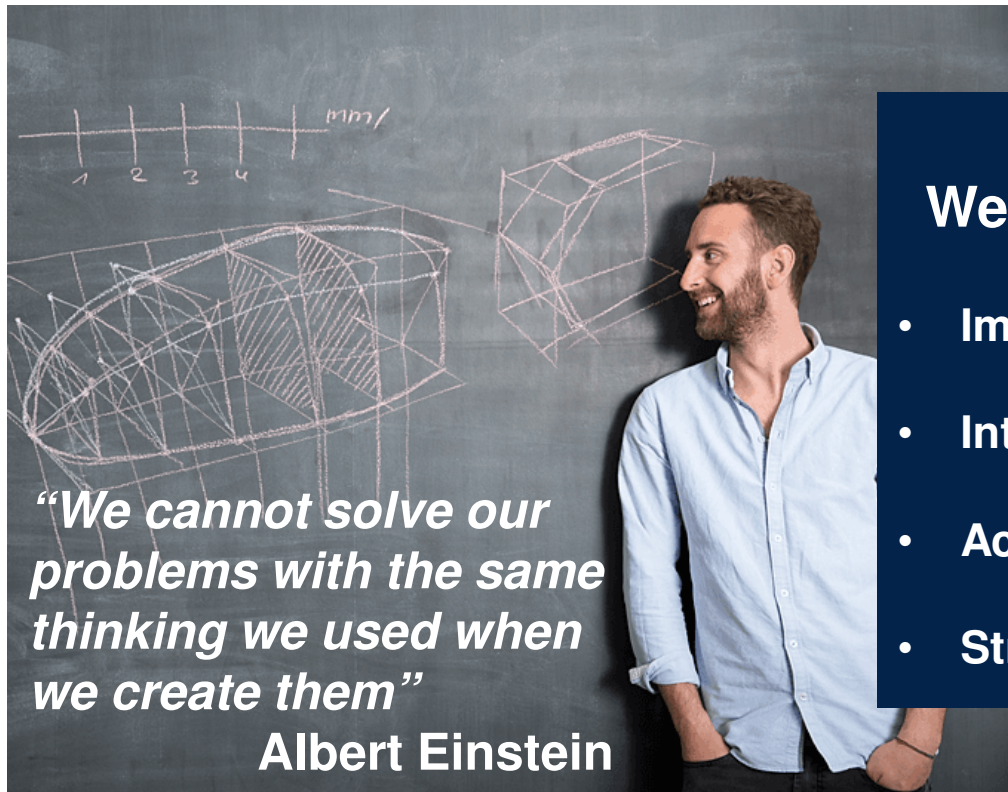
# STM32Cube Ecosystem

Tools and software supporting you during all your design steps

|                       |                                                                                                                |                                                                                               |
|-----------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Evaluation            |                               | Large and affordable selection of evaluation kits                                             |
| Getting Started       | <br>STM32<br>CubeMX           | Efficient and intuitive MCU configuration with automatic code generation                      |
| Development and Debug | <br>STM32<br>CubeMCUPackages  | Optimized drivers and middleware, full access to hardware features, production-ready examples |
|                       | <br>STM32<br>CubeExpansion    | Extended selection of optimized middleware and part drivers                                   |
|                       | <br>STM32<br>CubeIDE        | Eclipse & VSCode, C/C++ code editing, GCC & LLVM build, advanced debugging                    |
| Programming           | <br>STM32<br>CubeProgrammer | Efficient and secure code and option bytes programming                                        |
| Monitoring            | <br>STM32<br>CubeMonitor    | Graphical and non-intrusive code monitoring                                                   |

# STM32Cube Ecosystem

STM32Cube Ecosystem is evolving to help you even more



**We are upgrading our STM32Cube Ecosystem to:**

- Implement developers' expected enhancements and fixes
- Introduce more advanced features and deploy them at scale
- Accelerate our reactivity and release cycles
- Strengthen our foundation to support richer STM32 MCUs roadmap

# STM32Cube Ecosystem

**STM32Cube Ecosystem is evolving to help you even more**

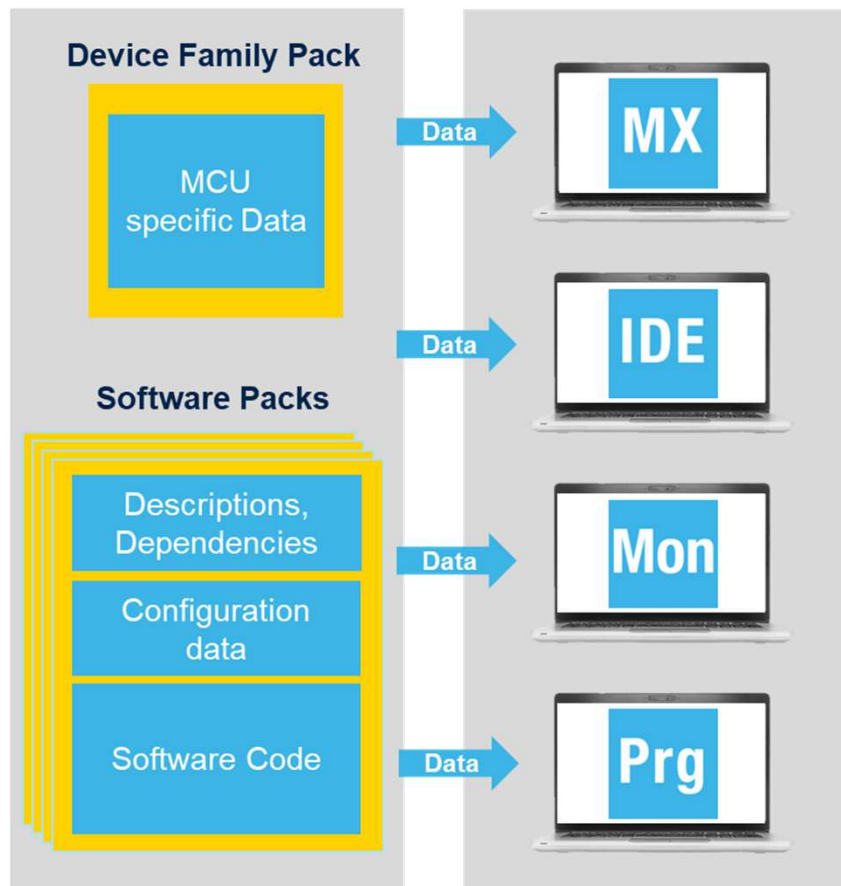


## **Our new architecture principals**

- **Decoupled embedded software and tools**
  - **Modular and configurable embedded software components**
  - **Data-centric tools, STM32 MCUs agnostic**
- **Inter-operable tools instead of tools integration**

# STM32Cube Ecosystem

## STM32Cube Ecosystem new architecture principals



- **Device Family Pack (DFP):**
  - Data Describing each MCU (Pinout, clock, SVD, Flash, startup)
- **Modular software components (Software Packs)**
  - 1 driver pack for each series
  - 1 pack for each Middleware, example, utility and external part drivers
- **Open-CMSIS-PACK format for each pack description**
  - Data describing software licenses, versions, dependencies and configuration
- **Hardware-Independent tools**
  - Software tools use DFP and Software Packs data as standard input
  - Capacity to automatically fetch and add dependencies.

# STM32Cube Ecosystem

STM32Cube Ecosystem is evolving to help you even more

|                       |                                                                                     |                                                                               |
|-----------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Evaluation            |    | Faster start from boards, faster access to examples and demos                 |
| Getting Started       |    | Enhanced configuration, code preview, flexible code structure, easier porting |
| Development and Debug |    | Higher Modularity, lower footprint and higher quality                         |
|                       |   | Simplified extensions                                                         |
|                       |  | VSCode as new framework, enhanced edit, build and debug                       |
| Programming           |  | Increased speed and extended probes support                                   |
| Monitoring            |  | More specialized monitoring                                                   |

# STM32CubeMCU Packages Evolution

HAL drivers are upgraded, LL drivers are unchanged

HAL → HAL2.x

- **Optimized footprint:** up to 59%\* lower flash size
  - **Improved API:** simplified APIs, faster execution, added missing functions
  - **Improved portability:** compatibility across STM32 series
  - **Enhanced quality:** IEC61508/SC2 and MISRA C compliant, statical analysis
  - **HAL2 built on top of LL:** HAL2 now calling LL APIs
- **Saving memory and cost**
  - **Shorter design, higher perf.**
  - **Accelerated migration**
  - **More reuse, Safety ready code**
  - **Easier and faster optimization**

\*measured on RCC on STM32U5

# STM32CubeMCU Packages Evolution

Our embedded software packages are reworked

STM32  
CubeMCU Packages



STM32  
CubeMCU Packages V2

- **Modular software components:** unitary, OPEN-CMSIS-PACK based
- **More examples:** reference use of all STM32 MCU features
- **Improved example's structure** made boards and IDEs agnostic
- **Enhanced BSP architecture:** based on part driver's layer

- ➔ • **More flexibility, easier update**
- ➔ • **Faster development and debug**
- ➔ • **Easier reuse in end application**
- ➔ • **Easier reuse in end application**

# STM32CubeMX Evolution

Faster development, alignment with embedded SW evolution



- **Data centric:** MCU agnostic, using CMSIS-Packs components as input
  - **Upgraded Configuration:** Enhanced pinout, clock and IPs settings, more flexible code generation
  - **Porting support:** Supporting inter and cross series migration
  - **New Code preview:** Live view of generated code with easy copy
  - **Improved UX:** New views and highlights, simplified journeys
  - **Enhanced CLI:** full access to functionalities through command line
- **Flexibility on tool used versions**
  - **Fast and optimized configuration**
  - **Accelerated porting**
  - **Easier code access and reuse**
  - **Faster and easier configuration**
  - **Automated CI/CD support**

# STM32 IDE Evolution

Faster development, alignment with embedded SW evolution



- **Data Centric:** MCU agnostic, using CMSIS-Packs components as input
  - **VSCode based, decoupled from MX:** lighter, faster, more evolutive
  - **CMake based, GCC and LLVM support:** decoupling projects and build
  - **Modular features addition:** flexible install with 'Bundle manager' feature
  - **Enhanced UX:** simple journeys, internationalization support
  - **Separated CLI:** full access to functionalities through command line
- **Flexibility on tool used versions**
  - **Faster install and development**
  - **more flexibility, faster build**
  - **Flexible install, disk space saving**
  - **Easier to use tool**
  - **Automated CI/CD support**

# Our technology starts with You



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